

Vedant Shinde

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EDUCATION

University at Buffalo

MS in Data Science

Coursework: Machine Learning, Statistical Learning, Probability, Data Models and Query Language, Reinforcement Learning

Buffalo, USA

Aug 2025 - Dec 2026

Vasantdada Patil Pratisthan's College of Engineering and Visual Arts

BE in Information Technology

Mumbai, India

Jun 2017 - Jun 2021

EXPERIENCE

Morgan Stanley

Senior Associate - SRE Automation

Mumbai, India

Oct 2021 - Jul 2025

- Engineered anomaly and volume deviation detection models for MQ flows using statistical methods (**z-score**, **standard normal**), integrated with Prometheus and Grafana, reducing root-cause analysis time by 40% and preventing 10+ potential failures per month.
- Designed and presented a one-stop "**Definition of Done**" dashboard for 15+ OTC and ETD mandates, enabling senior management to track **100,000+ daily EOD transactions**, improving decision-making speed by 40%.
- Applied **machine learning** models (**Isolation Forest**, **GMM**) for log-stream anomaly detection, improving incident detection accuracy by **25%** and reducing manual triage effort.
- Automated alert-parsing pipeline querying a **data warehouse** with **regex-based asset detection**, auto-generating **Jira tickets** routed to relevant squads, reducing operational TOIL by **30%**.
- Built a batched **log telemetry streaming pipeline** using **Prometheus Pushgateway** and **PromQL**, enabling traceability across **50+ Java microservices** and reducing mean-time-to-diagnose (MTTD) by **10%**.

PROJECTS

Supply Chain Data Pipeline (2026)



- Architected an end-to-end **data engineering pipeline** ingesting **600,000+ supply chain records** into **PostgreSQL** using modular **Python**, **Pandas**, and **SQLAlchemy** ingestion workflows.
- Modeled a normalized **3NF OLTP database** from raw CSV data, enforcing relational integrity and eliminating redundancy through SQL-based schema design.
- Created a **dbt** transformation layer to convert normalized operational data into an analytics-ready **star schema**, producing reusable **fact** and **dimension** models for supply chain analytics.
- Integrated automated **dbt data quality tests**, version-controlled SQL transformations, and deployed an interactive **Streamlit** dashboard on **Render**.

WorldLens Analytics Platform (2026)



- Designed an end-to-end **ELT pipeline** ingesting socioeconomic indicators from the **World Bank API** into **Snowflake** for **260+ countries** across multiple decades using configurable Python ingestion workflows.
- Enabled **idempotent loading**, configurable historical back fills, and ingestion tracking to ensure reliable, repeatable, duplicate-free data ingestion.
- Developed reusable **dbt** models spanning staging, dimensions, facts, and marts to support country-level analytical reporting using dimensional modeling best practices.
- Automated **dbt data quality tests** and SQL transformations, delivering trusted analytical datasets powering interactive cross-country trend dashboards.

HiveBlot - 2nd Place - UCSF Quantitative Biosciences Institute (QBI) Hackathon (2026)

- Designed a curated **One Big Table (OBT)** data model by transforming OCR and LLM-extracted **Western blot experimental evidence** into a unified schema, enabling rapid evidence retrieval across **15+ biomedical research papers**.
- Orchestrated a modular extraction workflow integrating **OCR**, computer vision, and LLM-based information extraction to identify proteins, experimental conditions, biological outcomes, and associated figures across **15+ biomedical research papers**.
- Delivered a centralized evidence indexing system enabling researchers to retrieve experimental findings in seconds instead of manually reviewing **15+ papers** (saving an estimated **2-3 hours** per literature search), with traceability to original figures.

Online Retail Analysis (2025)



- Performed feature engineering (RFM analysis, one-hot encoding) and data preprocessing on **500k+ transactions**, improving model performance and training efficiency.
- Evaluated **Logistic Regression**, **XGBoost**, and **Random Forest**; selected **Random Forest** based on highest **F1-score** and superior robustness to class imbalance.
- Deployed the best-performing model via **FastAPI** for real-time inference, containerized with **Docker** and hosted on **Render**, enabling scalable predictions for high-value customer targeting.

SKILLS

- Programming:** Python, SQL, R
- Data Engineering:** ETL/ELT, Data Pipelines, Apache Airflow, dbt, Data Modeling, Dimensional Modeling, Star Schema, OLTP/OLAP, Data Warehousing
- Databases & Lakehouse:** PostgreSQL, Snowflake, Databricks, PySpark, Delta Lake, Unity Catalog
- Frameworks & Libraries:** Pandas, NumPy, SQLAlchemy, FastAPI, Docker, Streamlit, Scikit-Learn, LLM, Flask, Matplotlib
- Cloud & Tools:** GCP, Git, AWS, Jenkins, Prometheus, Grafana, MLflow, Statistical Analysis, Render, Linux